

A bound for the number of maximum modulus points

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

For a non-monomial entire function f , let $\nu_f(r)$ be the number of points on $|z| = r$ at which $|f|$ attains its maximum. Write $f(z) = z^m(c_0 + c_k z^k + O(z^{k+1}))$, where $c_0 c_k \neq 0$ and $k \geq 1$. We prove that $\nu_f(r) \leq 2k$ outside a countable set of radii. In particular, $\nu_f(r)$ cannot tend to infinity, answering the second question of Erdős on maximum modulus points. The proof uses continuation of implicit functions and growth in direct tracts to study the complex curve $A(z) = \overline{A(\overline{w})}$, where $A = zf'/f$. On the locus $zw \neq 0$ and $A(z) \notin \{m, \infty\}$, the map $(z, w) \mapsto (zw, A(z))$ is finite and proper. Counting its fibres near the coordinate axes gives the bound. No growth assumption on f is required.

1 Introduction

For a nonzero entire function f , put

$$M(r, f) = \max_{|z|=r} |f(z)|, \quad \nu_f(r) = \#\{z : |z| = r, |f(z)| = M(r, f)\}.$$

If f is not a monomial, then $\nu_f(r)$ is finite for every $r > 0$. Erdős asked whether a single such function can satisfy

$$\limsup_{r \rightarrow \infty} \nu_f(r) = \infty, \quad \text{and whether it can satisfy} \quad \liminf_{r \rightarrow \infty} \nu_f(r) = \infty;$$

see [2, 4] for the problem and its history. Herzog and Piranian [6] answered the first question affirmatively. Pardo-Simón and Sixsmith [8] obtained such an example of finite order. Glücksam and Pardo-Simón [4] constructed an approximate analogue of the second property, with many separated arcs on which the modulus is close to its maximum.

We prove that the answer to the second question is negative.

Theorem 1.1. *Let f be a nonzero entire function that is not a monomial, and write*

$$f(z) = z^m(c_0 + c_k z^k + O(z^{k+1})), \quad m \geq 0, \quad k \geq 1, \quad c_0 c_k \neq 0. \quad (1.1)$$

There is a countable set $E \subset (0, \infty)$ such that

$$\nu_f(r) \leq 2k \quad (r \notin E). \quad (1.2)$$

Consequently,

$$\liminf_{r \rightarrow \infty} \nu_f(r) \leq 2k < \infty.$$

The integer k is the difference between the degrees of the first two nonzero terms of f . The factor 2 is attained by an explicit example given in Remark 5.1.

To explain the argument, set $A = zf'/f$. At any radius where $\log M(r, f)$ is differentiable with respect to $\log r$, all maximum modulus points have the same real value of A . Thus they give points $(z, \bar{z})$ of the complex curve $A(z) = \overline{A(\bar{w})}$, with the same values of both zw and $A(z)$. We study the fibres of the resulting map of curves. The main step is to show that every irreducible component contains points with $|zw|$ arbitrarily small. At such points one coordinate lies near the origin, where $A - m$ has local degree k . Properness then carries this local count to every component of the image, outside a discrete set.

2 Two analytic facts

We use the following consequence of the continuation theorem for implicit functions defined by an entire relation.

Lemma 2.1 (Julia). *Let F be entire on $\mathbb{C}^2$, and suppose that*

$$F(a, b) = 0, \quad F_w(a, b) \neq 0.$$

The projection to the z -plane of the maximal analytic continuation of the implicit germ $w = w(z)$ through (a, b) is dense in $\mathbb{C}$.

Julia [7, Section III] proves that this projection cannot omit a continuum. The stronger Iversen continuation property, due to Stoilow, is stated in [3]: the germ can be continued along an arbitrarily small perturbation of any prescribed path starting at a . Either statement gives Lemma 2.1.

The second fact concerns growth in a direct tract. In the formulation of Bergweiler, Rippon, and Stallard [1, Definition 2.1], an unbounded domain $D \subset \mathbb{C}$, with unbounded complement and piecewise smooth boundary, is a *direct tract* of Q if Q is holomorphic in D , continuous on its closure, and

$$|Q| = a \quad \text{on } \partial D, \quad |Q| > a \quad \text{in } D$$

for some $a > 0$. No condition is imposed on Q outside $\overline{D}$.

Lemma 2.2 (Fuchs; see [1, Theorem 2.1]). *Let D be a direct tract of Q , with boundary value a , and put*

$$u(w) = \begin{cases} \log(|Q(w)|/a), & w \in D, \\ 0, & w \notin D. \end{cases}$$

Then u is a continuous subharmonic function on $\mathbb{C}$, and

$$\frac{\max_{|w|=R} u(w)}{\log R} \longrightarrow \infty \quad (R \rightarrow \infty). \quad (2.1)$$

The limit in (2.1) has no exceptional radii; see [1, Section 9] for a proof.

3 Small products on the correspondence

Fix f, m, k as in Theorem 1.1. Write $f = z^m h$, where $h(0) \neq 0$, and define the meromorphic functions

$$A(z) = \frac{zf'(z)}{f(z)} = m + \frac{zh'(z)}{h(z)}, \quad B(w) = A^\#(w) := \overline{A(\bar{w})}. \quad (3.1)$$

The removable singularity of A at zero is filled in. The function A is nonconstant, since otherwise f would be a monomial. Moreover, (1.1) gives

$$A(z) - m = \frac{kc_k}{c_0} z^k + O(z^{k+1}).$$

In particular, $k = \text{ord}_0(A - m)$.

There is a holomorphic local coordinate χ at zero such that

$$\chi(0) = 0, \quad \chi'(0) \neq 0, \quad A(z) - m = \chi(z)^k. \quad (3.2)$$

Choose $\delta \in (0, 1)$ small enough that

$$V = \chi^{-1}\{\xi : |\xi| < \delta^{1/k}\}$$

has compact closure in the coordinate neighbourhood. For every $|\lambda - m| < \delta$, the equation $A(z) = \lambda$ has exactly k roots in V , counted with multiplicity. They are distinct when $\lambda \neq m$, and there is a constant C such that all of them satisfy

$$|z| \leq C|\lambda - m|^{1/k}. \quad (3.3)$$

The corresponding statements hold for B on $V^\# = \{\bar{z} : z \in V\}$.

We also require δ to differ from the modulus of every finite, nonzero critical value of $B - m$. This excludes at most countably many choices. The level set $\{|B - m| = \delta\}$ therefore consists of smooth curves that are locally finite in the plane.

Choose entire functions p, q with no common zero such that $A = p/q$ and $q(0) \neq 0$. Set

$$F(z, w) = p(z)q^\#(w) - p^\#(w)q(z), \quad \mathcal{X} = \{(z, w) \in \mathbb{C}^2 : F(z, w) = 0\}. \quad (3.4)$$

All analytic sets in this paper are given their reduced structure. Away from poles, the equation in (3.4) is $A(z) = B(w)$. Since p, q have no common zero and A is nonconstant, $\mathcal{X}$ has no horizontal or vertical irreducible component. At points with finite common value,

$$F_w(z, w) = -q(z)q^\#(w)B'(w).$$

It follows that $F_w \neq 0$ at a general point of each irreducible component: the coordinate w is nonconstant there, whereas the zeros of B' are discrete.

Lemma 3.1. *Every irreducible component X of $\mathcal{X}$ contains a sequence (z_j, w_j) such that*

$$z_j w_j \neq 0, \quad A(z_j) = B(w_j) \in \mathbb{C} \setminus \{m\}, \quad |z_j w_j| \longrightarrow 0. \quad (3.5)$$

Proof. Start with a regular implicit germ on X at which $F_w \neq 0$. Its analytic continuations remain in X . Lemma 2.1 gives a point $(z_0, w_0) \in X$ with

$$z_0 \in V \setminus \{0\}, \quad 0 < |A(z_0) - m| < \delta.$$

Let U be the connected component of $\{|B - m| < \delta\}$ containing w_0 . In $V \times U$, consider the analytic curve

$$\chi(z)^k = B(w) - m. \quad (3.6)$$

Its projection to U is finite and proper. Indeed, over a compact subset of U , the right-hand side remains in a compact subset of $\{\xi : |\xi| < \delta\}$, so all its k -th roots remain in a compact subset of the coordinate disc.

Since $A'(z_0) \neq 0$, the curve (3.6) is smooth at (z_0, w_0) . Let Y be the irreducible component containing its unique local branch there. Analytic continuation gives $Y \subset X$. The projection $Y \rightarrow U$ is still finite and proper, and it is nonconstant. Its image is both open and closed in U , so it is surjective. We may therefore choose, for any $w \in U$, a point $(z, w) \in Y$ satisfying (3.3).

If $B(b) = m$ for some $b \in U$, choose $w_j \rightarrow b$ in U , with $w_j \neq 0$ and $B(w_j) \neq m$, and lift these points to Y . Equation (3.3) gives $z_j \rightarrow 0$, and hence $z_j w_j \rightarrow 0$. This proves the lemma in this case.

Suppose now that $B - m$ has no zero in U , and put

$$Q(w) = \frac{\delta}{B(w) - m}. \quad (3.7)$$

Every finite boundary point of U satisfies $|B - m| = \delta$. In particular, no such point is a zero or pole of $B - m$. Thus Q is holomorphic in U , continuous on $\bar{U}$, and satisfies

$$|Q| > 1 \quad \text{in } U, \quad |Q| = 1 \quad \text{on } \partial U.$$

The maximum principle shows that U is unbounded.

We claim that its complement is also unbounded. If f has a zero other than the origin, then for every sufficiently large R whose circle avoids the zeros of f , the argument principle gives

$$\frac{1}{2\pi} \int_0^{2\pi} (B(Re^{i\theta}) - m) d\theta = n(R, f) - m \geq 1, \quad (3.8)$$

where $n(R, f)$ counts zeros in $|z| < R$, with multiplicity. Here $B(w) = w(f^\#)'(w)/f^\#(w)$, and f and $f^\#$ have the same zero counts on centred discs. Since $\delta < 1$, such a circle cannot be contained in $\{|B - m| < \delta\}$. A circle containing a pole of B is not contained in that set either. If f has no zero other than possibly the origin, then $B - m$ is a nonconstant entire function. Its maximum modulus eventually exceeds δ , giving the same conclusion for all large circles. In either case, $\mathbb{C} \setminus U$ is unbounded.

Consequently, U is a direct tract of Q with boundary value one. Define $u = \log |Q|$ in U , and $u = 0$ outside U . For each sufficiently large R , choose w_R on $|w| = R$ at which u attains its maximum. By Lemma 2.2, this maximum is positive and

$$\frac{u(w_R)}{\log R} \rightarrow \infty.$$

In particular, $w_R \in U$. Choose a lift $(z_R, w_R) \in Y$. Equations (3.3) and (3.7) yield

$$|z_R w_R| \leq C \delta^{1/k} R \exp\left(-\frac{u(w_R)}{k}\right) \rightarrow 0. \quad (3.9)$$

Since $B - m$ has no zero in U , these lifts have nonzero coordinates and common value different from m . They satisfy (3.5). $\square$

4 A finite map of analytic curves

Remove the zero coordinates, poles, and common value m , and set

$$\begin{aligned} \mathcal{X}^* &= \{(z, w) \in \mathcal{X} : zw \neq 0, A(z) = B(w) \in \mathbb{C} \setminus \{m\}\}, \\ \mathcal{B} &= \mathbb{C}^* \times (\mathbb{C} \setminus \{m\}). \end{aligned}$$

Consider the holomorphic map

$$\Phi : \mathcal{X}^* \rightarrow \mathcal{B}, \quad \Phi(z, w) = (zw, A(z)) = (s, \lambda). \quad (4.1)$$

Lemma 4.1. *The map Φ is proper and has finite fibres.*

Proof. Suppose that (s, λ) remains in a compact subset of $\mathcal{B}$. Then $|s|$ is bounded above and away from zero, while λ is bounded and stays a positive distance from m . If a sequence of preimages had $z \rightarrow \infty$, then $w = s/z \rightarrow 0$, and therefore $\lambda = B(w) \rightarrow m$, a contradiction. If $z \rightarrow 0$, then $\lambda = A(z) \rightarrow m$, again a contradiction. The same arguments apply with z and w interchanged. Neither coordinate can approach a pole, since λ is bounded. It follows that the inverse image of the compact set is compact in $\mathcal{X}^*$.

For fixed (s, λ) , the possible z -coordinates form a compact subset of the discrete zero set of $A - \lambda$. There are therefore only finitely many of them, and each determines $w = s/z$ uniquely. $\square$

Proposition 4.2. *The image $\Gamma = \Phi(\mathcal{X}^*)$ is a closed analytic curve in $\mathcal{B}$. There is a locally finite set $T \subset \Gamma$ such that*

$$\#\Phi^{-1}(s, \lambda) \leq 2k \quad ((s, \lambda) \in \Gamma \setminus T). \quad (4.2)$$

Proof. The proper mapping theorem and Lemma 4.1 imply that Γ is a closed analytic curve; see [5, Chapter IV, Section 1, Corollary 2] for the finite-fibre projection theorem. Let T consist of the singular points of Γ , the images of singular points of $\mathcal{X}^*$, and the critical values of Φ on the regular part of $\mathcal{X}^*$. These sets are locally finite. For the critical points this follows by using local parameters on the normalizations of the source branches: $\lambda = A(z)$ is nonconstant on each component, so its derivative is not identically zero. Properness ensures that only finitely many such branches need be considered over a compact neighbourhood in the target.

For each irreducible component Γ_i of Γ , the restriction over $\Gamma_i \setminus T$ is a finite unramified covering. The number of points in a fibre is consequently locally constant there. The regular part of an irreducible curve is connected, and remains connected after deletion of a locally finite set; see [5, Chapter IV, Section 3, Corollary 1]. Thus there is an integer d_i such that

$$\#\Phi^{-1}(s, \lambda) = d_i \quad ((s, \lambda) \in \Gamma_i \setminus T). \quad (4.3)$$

The integer d_i counts all preimages, including those belonging to different components of $\mathcal{X}^*$.

Choose $\eta > 0$ so that the disc $\{|z| < \eta\}$ is contained in both V and $V^\#$. If $0 < |s| < \eta^2$, every preimage satisfies

$$\min\{|z|, |w|\} < \eta.$$

There are at most k preimages with $|z| < \eta$: the existence of such a coordinate implies $|\lambda - m| < \delta$, and the local root count in Section 3 applies. Each root determines the other coordinate from $w = s/z$. Likewise there are at most k preimages with $|w| < \eta$. Hence

$$\#\Phi^{-1}(s, \lambda) \leq 2k \quad (0 < |s| < \eta^2). \quad (4.4)$$

Every irreducible component of $\mathcal{X}^*$ is obtained from one of $\mathcal{X}$ by deleting a discrete set. Its image under the finite proper map Φ is an irreducible component of Γ , and every component of Γ arises in this way. Lemma 3.1 therefore shows that each Γ_i meets $\{0 < |s| < \eta^2\}$. This intersection is a nonempty open subset of a curve, so it is not contained in T . Evaluating (4.3) at a point of this intersection and using (4.4) gives $d_i \leq 2k$. Equation (4.2) follows for every component. $\square$

5 Maximum modulus points

Proof of Theorem 1.1. Set

$$v(x) = \log M(e^x, f), \quad x \in \mathbb{R}.$$

Hadamard's three-circles theorem makes v convex. Its set N of nondifferentiability points is therefore at most countable. Suppose that $x \notin N$, let $r = e^x$, and let $z = re^{i\theta}$ be a maximum modulus point. Since $f(z) \neq 0$, angular differentiation gives

$$0 = \frac{\partial}{\partial \theta} \log |f(re^{i\theta})| = -\operatorname{Im} A(z). \quad (5.1)$$

For this fixed θ , the function $t \mapsto \log |f(e^{t+i\theta})|$ lies below $v(t)$ and agrees with it at $t = x$. Differentiating at this contact point yields

$$\operatorname{Re} A(z) = v'(x). \quad (5.2)$$

Thus all maximum modulus points on the circle have the same real value $A(z) = \lambda := v'(x)$.

In fact $\lambda > m$. To see this, write

$$b(x) = v(x) - mx = \log M(e^x, h).$$

The function b is convex and nondecreasing. Since h is nonconstant, the maximum principle gives

$$b(x) > \log |h(0)| = \lim_{t \rightarrow -\infty} b(t) \quad (x \in \mathbb{R}).$$

If $b'(x) = 0$ at a differentiability point, convexity and monotonicity would force b to be constant on $(-\infty, x]$, a contradiction. Hence $\lambda = m + b'(x) > m$.

For every maximum modulus point z on $|z| = r$, we now have

$$(z, \bar{z}) \in \mathcal{X}^*, \quad \Phi(z, \bar{z}) = (r^2, \lambda). \quad (5.3)$$

Distinct maximum points give distinct points of this fibre. If $(r^2, \lambda) \notin T$, Proposition 4.2 gives $\nu_f(r) \leq 2k$.

The locally finite set T is countable, because $\mathcal{B}$ is second countable. Define

$$E = \{e^x : x \in N\} \cup \{\sqrt{s} : (s, \lambda) \in T, s \in (0, \infty)\}.$$

This is a countable set of positive radii, and (1.2) holds for every $r \notin E$. Since every interval (R, ∞) contains radii outside E , the asserted bound on the lower limit follows. $\square$

Remark 5.1. For $k \geq 1$, the function $f(z) = \exp(2z^k - z^{2k})$ has first two nonzero Taylor terms $1 + 2z^k$. Put $R = r^k$ and $\phi = k\theta$. Then

$$\log |f(re^{i\theta})| = R^2 + \frac{1}{2} - 2R^2 \left(\cos \phi - \frac{1}{2R} \right)^2.$$

For $R > 1/2$, the maximum is attained precisely when $\cos(k\theta) = 1/(2R)$, giving $2k$ distinct points. Thus the bound in Theorem 1.1 is sharp for every k .

AI tool disclosure

AI tools were used substantially in developing this work. Earlier AI-assisted explorations supplied the logarithmic-derivative formulation used in Section 3 and the example in Remark 5.1. OpenAI's GPT-6, accessed through Codex, developed the argument in Section 3 combining implicit-function continuation with growth in a direct tract, the finite-map bound in Section 4, and their application to maximum modulus points in Section 5. GPT-6 was also used for literature retrieval, proof checking, and preparation of the English text and L^AT_EX source. Responsibility for the mathematical content and the accuracy of the references rests with the author.

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